

Can robots elicit empathy? The effects of social robots' appearance on emotional contagion

Wenjing Yang, Yunhui Xie^{*}

School of Economics and Management, Beihang University, Beijing, China

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ABSTRACT

The increasing integration of robots as service providers or companions in daily life has prompted extensive research on the emotional aspects of human-robot interaction (HRI). Emotional contagion, a crucial factor influencing user experience in HRI, has gained considerable attention. However, limited research explores the influence of anthropomorphism and gender differences in robot modeling on emotional contagion in HRI, leaving a gap in reference guidance for social robot design and application. To address this, we investigate human-robot interactions in service scenarios, analyzing the impact of robot appearance, anthropomorphism, and gender on the transmission of positive and negative emotions. The experimental findings highlight the significant role of robot gender in influencing emotional contagion in HRI, while revealing the interaction effect between robot gender and anthropomorphism on emotional contagion. This interdisciplinary study provides empirical evidence that enriches research on the emotional aspects of HRI and contributes to more informed design of user experiences.

1. Introduction

Social robots, employing fully or partially automated technologies to co-create value with humans through their social functions, have become an increasingly important element in the modern service industry (Hagele, 2016). Social robots hold growing economic and societal significance and have been widely applied in various domains such as education, home environments, and psychological therapy (Share, 2018). For instance, Pepper, developed by Softbank Robotics, has found extensive application in retail and food service industries (Abildgaard & Scharfe, 2012). DJI RoboMaster S1 by DJI is widely used in STEM education (Sanders Jr, 2021), and Lovot by Groove X offers companionship and emotional support services (Yoshida, et al., 2022). It is noteworthy that emotions have emerged as a crucial factor influencing the user experience in human-robot interaction (HRI) (Rincon, Costa, Novais, Julian, & Carrascosa, 2019; Shank, Graves, Gott, Gamez, & Rodriguez, 2019) as robots gradually serve humans with social attributes.

In the field of human responses to robot emotions, empathy plays a significant role in HRI, referring to the robot's ability to recognize, understand, and respond to human emotions, fostering a sense of connection and rapport with users (Birmingham, Perez, & Matarić, 2022). Actually, "human responses to robot emotions" is one important

aspect in psychological research on robotics with the other three including (1) the generation of artificial robotic emotions (stimulus), (2) human recognition of robotic artificial emotions (organism), and (3) other contingencies that affect emotions (Stock-Homburg, 2022). Studies have explored various techniques and algorithms to imbue robots with empathetic capabilities, enabling them to perceive and appropriately react to users' emotional states (Bagheri, Roesler, Cao, & Vanderborght, 2021). In the study of empathy, emotional contagion in HRI emphasizes the emotional state of robots or individuals transmitted to the other one through nonverbal or verbal forms (Hatfield, Cacioppo, & Rapson, 1993).

The issue of emotional contagion in HRI is increasingly important with the widespread use of artificial intelligence (AI) tools, such as chatbots or voice assistants, as a means to facilitate or support business interactions and transactions (Chuah & Yu, 2021; Houston III, Grandey, & Sawyer, 2018; Matsui & Yamada, 2019; Otterbring & Lu, 2018). Emotional contagion in social interactions is a crucial characteristic for understanding emotional states (Herrando & Constantinides, 2021), while it helps the understanding of individuals' emotional states in HRI. Specifically, individuals in HRI demonstrate behavioral, attentional, and affective synchrony in response to emotional contagion (Hatfield, Cacioppo, & Rapson, 1992; Prochazkova & Kret, 2017), and this

^{*} Corresponding author. Beihang University, No. 37 Xueyuan Road, Haidian District, Beijing, 100191, China.

E-mail address: xieyh@buaa.edu.cn (Y. Xie).

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synchrony significantly enhances the user's HRI experience (Hegel, Spexard, Wrede, Horstmann, & Vogt, 2006). Due to the tendency of individuals to synchronize their emotional states with those perceived during social interactions, studying emotional contagion in HRI is of great significance. ADDIN EN.CITE ADDIN EN.CITE.DATA (Ekman, Levenson, & Friesen, 1983; Hatfield, Rapson, & Le, 2011).

While existing literature primarily examines the influence of robot actions or facial expressions on emotional contagion (Casso, Li, Nazir, & Delevoye-Turrell, 2022; Stock, 2016; Xu, Broekens, Hindriks, & Neerincx, 2015), limited attention has been given to the impact of robot appearances on emotional contagion. Studies focusing on virtual robot appearances have predominantly explored the effect of anthropomorphism on a single emotion. Specifically, a concrete agent can infect a positive emotion better than a cartoon-like agent (Matsui & Yamada, 2016). The studies mentioned above have shed light on the relationship between robot actions, facial expressions, and emotional contagion. However, a comprehensive exploration of the broader spectrum of emotions and their intricate interplay with various aspects of robot appearance remains largely uncharted. Understanding more non-verbal cues provided by robots, such as their body shape, perceived gender, clothing, etc., remains an important direction for future exploration (Urakami & Seaborn, 2023). Therefore, this study aims to specifically investigate the effects and mechanisms of social robot anthropomorphism and gender on the contagion of both positive and negative emotions in HRI.

This interdisciplinary research marks a significant step in advancing our understanding of emotional contagion in social robots, imparting fresh momentum for further advancements in the field. By shedding light on the intricate dynamics of emotional exchange between humans and robots, this study becomes a catalyst for stimulating vibrant discussions and innovative experience design within the realm of emotional aspects in Human-Robot Interaction (HRI) and the domain of social robotics. The insights garnered from this research hold the potential to shape the future development of social robots, enriching their emotional intelligence and facilitating more authentic and meaningful human-robot interactions.

In this paper, we (i) developed a theoretical model of the impact mechanisms of social robots on emotional contagion to users in Section 2. Subsequently, (ii) we conducted between-group experiments to investigate the effects of social robot anthropomorphism and gender on the emotional contagion effect, and within-group experiments to observe changes in users' positive and negative emotions in Section 3. Finally, (iii) based on the experimental results in Section 4, we revised our theoretical model and provided recommendations to relevant stakeholders for further improving the design of social robot appearances in Section 5.

2. Literature review and hypothesis development

2.1. The critical role of emotions in interpersonal relationships and social interactions

Emotions are a cornerstone in human social interactions, playing a pivotal role in the formation and maintenance of interpersonal relationships. Scholarly research indicates that emotions not only sculpt our decision-making processes but also profoundly influence social interactions and group dynamics (Keltner & Haidt, 1999). In human social exchanges, the expression and recognition of emotions are crucial mechanisms for establishing trust, fostering collaboration, and resolving conflicts (Fiske, Cuddy, & Glick, 2007). As social robotics advance, the role of emotions in Human-Robot Interaction (HRI) gains increasing significance. Studies demonstrate that if social robots can appropriately display and recognize emotions, they become more effective in establishing trust and connection with users (Reeves & Nass, 1996). This highlights the potential of mimicking human emotional communication patterns in HRI. For instance, emotional contagion can affect the

emotional climate and performance of teams in organizational and team settings (Barsade, 2002). This finding provides a theoretical foundation for the study of emotional contagion in HRI.

Upon further examination, in HRI, emotional contagion is not only a phenomenon warranting deeper investigation but also a critical consideration in design. This pertains not just to the quality of the user experience but also to the effectiveness of social robots in various application contexts, such as education, healthcare, and customer service (Zhang, Yang, & Wang, 2022). When contemplating the design and functionality of social robots, applying these principles of human emotional dynamics to the human-machine interaction environment can aid in crafting robots that communicate and interact with humans more effectively.

2.2. Emotional contagion in human-robot interaction

In recent years, emotional contagion has become an important factor in enhancing user experience in HRI (Gong, 2007; Riek, Paul, & Robinson, 2010), consequently, the phenomenon of emotional contagion in HRI has garnered increasing attention and discussion within the academic community. Xu et al. (2015) demonstrated through experiments involving gesture interactions with a social robot that users' emotional states were influenced in the expected direction by the robot's emotions, highlighting the contagious effect of robot emotions on users. Stock (2016) indicated that emotional contagions in HRI, such as between frontline service robots and human customers, may occur through customers imitating the language and body expressions of the robot.

Various factors influence the effect of emotional contagion in HRI, such as the category of emotions conveyed by the robot to humans, robot characteristics, and robot actions. Stock (2016) revealed that the contagion effect of negative emotions in HRI may be more pronounced than that of positive emotions. Matsui and Yamada (2016) demonstrated that the level of anthropomorphism of virtual characters affects the contagion of emotions by comparing product recommendations by virtual characters with different levels of anthropomorphism. Nomura, Kanda, and Suzuki (2006) found that when individuals have negative impressions of the robot's appearance, they are less likely to be influenced by the robot's emotions and are also less willing to engage in emotional communication with the robot. The study by Song, Tao, and Luximon (2023) highlighted that integrating positive emotional expressions in robots enhances users' trust and emotional involvement. Casso et al. (2022) suggested that slower robot movements, which are perceived as more natural and knowledgeable, are more likely to facilitate emotional contagion.

2.3. The effects of robot anthropomorphism on emotional contagion

Anthropomorphism is a phenomenon that describes humans' tendency to perceive human-like shapes in their environment, which significantly influences their choices and beliefs (Zlotowski, Proudfoot, Yogeewaran, & Bartneck, 2015). Researchers consider two main forms of robot anthropomorphism: perceptual and functional anthropomorphism, referring to a visual resemblance to humans and behavioral mapping similarity, respectively (Abildgaard & Scharfe, 2012). This study primarily focuses on perceptual anthropomorphism.

The anthropomorphic appearances of social robots can shape users' emotional attitudes towards them, subsequently influencing the robots' capacity for emotional contagion. As highlighted by Bartneck, Kanda, Mubin, and Al Mahmud (2007), humans tend to perceive robots with anthropomorphic appearances as entities that exhibit lifelike features beyond mere machines, imbuing them with a higher sense of vitality. Deaux (1984) demonstrated that robot anthropomorphism can engender greater trust from individuals, as people are more inclined to believe that anthropomorphized robots can meet their expectations and goals. Bartneck, Kulić, Croft, and Zoghbi (2009) further reported that designing robots in anthropomorphic forms promotes emotional

resonance, making them more readily perceived as social companions by humans and enhancing interaction and communication with robots. Moreover, anthropomorphic design can reduce fear and resistance towards robots, thereby increasing acceptance and trust among people (Bartneck, Kanda, Ishiguro, & Hagita, 2007). As a result, people are more likely to perceive the conveyed emotions by anthropomorphic designed robots as authentic and genuine. The increased credibility of the robot's emotions may facilitate the process of emotional contagion, as users are more inclined to adopt and respond to emotions that they perceive as authentic and legitimate (Hennig-Thurau, Groth, Paul, & Gremler, 2006). This heightened emotional authenticity can further enhance emotional engagement between users and robots, potentially leading to a stronger emotional influence and contagion within the human-robot interaction.

Furthermore, the anthropomorphic design of social robots can enhance their emotional expression, thereby promoting individuals' understanding of these emotions. Specifically, the anthropomorphic design allows social robots to better mimic the emotional expressions and behaviors of human beings, which can be linked to understanding emotion and goal accuracy and used as a tool for rapport building (Stolzenwald & Bremner, 2017). On the other hand, Dittrich, Troscianko, Lea, and Morgan (1996) supported the hypothesis that dynamic features determine the observer's perception of emotions, which implies that robots making more accurate gestures also provide users with a more realistic experience of perceived emotions.

Another pathway through which the anthropomorphic features of social robots affect emotional contagion is via the establishment of trust. Research indicates that anthropomorphized appearances play a significant role in strengthening emotional connections between robots and users (Damiano & Dumouchel, 2018; Leite et al., 2013). Specifically, certain facial features of robots, such as the facial width-to-height ratio, are crucial in elevating perceived trustworthiness (Song & Luximon, 2021). These aspects of anthropomorphism, by fostering trust, pave the way for more profound levels of emotional contagion.

We posit that the enhancement of anthropomorphism in robots facilitates an improved capacity of robots to replicate human emotions, and also promotes more accurate perception and understanding of the robot's emotions by the user, consequently fostering strengthened emotional connections between humans and robots. This augmentation in emotional bonding potentially enables emotional contagion within the context of human-robot interaction. Based on the aforementioned argument, we propose the following hypothesis:

H1a. The level of robot anthropomorphism is significantly related to the contagion of positive emotions in HRI.

H1b. The level of robot anthropomorphism is significantly related to the contagion of negative emotions in HRI.

2.4. The effects of robot gender on emotional contagion

Gender features are one of the most crucial factors influencing the interaction between robots and humans in robot design (Tatsuya Nomura, 2017). Individuals are susceptible to cues from robot appearance, voice, and behavior, which may influence their interpretation of the robot's gender and subsequently lead to different behaviors based on whether they perceive themselves interacting with a male or female robot (Law, Chita-Tegmark, & Scheutz, 2021; Tatsuya Nomura, 2017). Nomura (2017) summarized that human factors, such as individuals' gender and education, affect their psychological and behavioral reactions toward robots. For example, Hone (2006) indicated that interactive agents capable of engaging with players, particularly female agents, can effectively reduce feelings of frustration and discouragement in challenging games. However, there is a lack of studies on the effects of robots' gender on emotional contagion which is one focus of this study.

First, human gender stereotypes can also be projected onto robots in certain contexts. Experimental results by Tay, Jung, and Park (2014)

demonstrated that participants preferred robots with stereotypically matching gender voices and names for occupations. The research findings of Hu, Pan, Luo, and Yu (2022) indicated that customers' willingness to use service robots is positively influenced by the alignment between occupational gender stereotypes and the gender of the service robot. Furthermore, the perceived personality of a robot can interact with gender stereotypes to affect user acceptance. Research by Reich-Stiebert et al. (2017) indicated that gender stereotypes related to learning did not apply to robots performing learning tasks that aligned with gender stereotypes. Moreover, a mismatch between the robot's gender and the gender typicality of the tasks increased students' willingness to engage with the robot in future learning activities (Reich-Stiebert & Eysel, 2017).

Second, there may be differences in emotional expressiveness between male and female images. Research findings of Becker, Kenrick, Neuberg, Blackwell, and Smith (2007) suggested that individuals exhibit faster and more accurate detection of anger expressions on male faces and happier expressions on female faces. In a study on gender and emotional expression, female angry faces were evaluated as more angry, while male happy faces were evaluated as more joyful (Hess, Adams, & Kleck, 2004). Brody (1997) pointed out that gender differences in emotional expressiveness were found to be culturally specific among Asian international, Asian-American, and European-American college students. This suggests that when robots of different genders express the same emotions, people's perceptions of the robots' emotional expressions may also be influenced by the gender of the robot, leading to potential biases.

As the aforementioned theoretical mechanisms, the perception of robot gender can indeed influence people's emotions during HRI. This study proposes the following hypothesis:

H2a. The gender of social robots influences the contagion of positive emotions in HRI.

H2b. The gender of social robots influences the contagion of negative emotions in HRI.

2.5. The interaction effect of robot anthropomorphism and gender

Scholars discussed the interaction effect between the level of robot anthropomorphism and gender. Seo (2022) highlighted that in comparison to male service robots, female service robots elicit a greater sense of happiness and higher satisfaction, and this effect is further magnified when the anthropomorphism level is high rather than low. The mechanism underlying this interaction effect might be that the resemblance of highly anthropomorphic service robots to humans in both appearance and behavior fosters individuals' stronger perception of robots' gender, and consequently intensifies gender-related effects, like gender stereotypes in emotional contagion. Specifically, a previous study noted that female smiles tend to convey greater happiness than male smiles (Sonnby-Borgström, Jönsson, & Svernnson, 2008), potentially contributing to the increased positive emotional experience of interacting with female robots in experimental contexts, compared with the male robots. In a similar vein, it is plausible that the effect of robots' gender on emotional contagion might vary at different levels of robots' anthropomorphism.

Therefore, we propose the following hypothesis.

H3a. The interaction of service robots' anthropomorphism and gender exerts an effect on the contagion of positive emotion.

H3b. The interaction of service robots' anthropomorphism and gender exerts an effect on the contagion of negative emotion.

2.6. Theoretical model

Based on the aforementioned theories in robot psychology and relevant research findings in HRI, the significance of robots' appearance

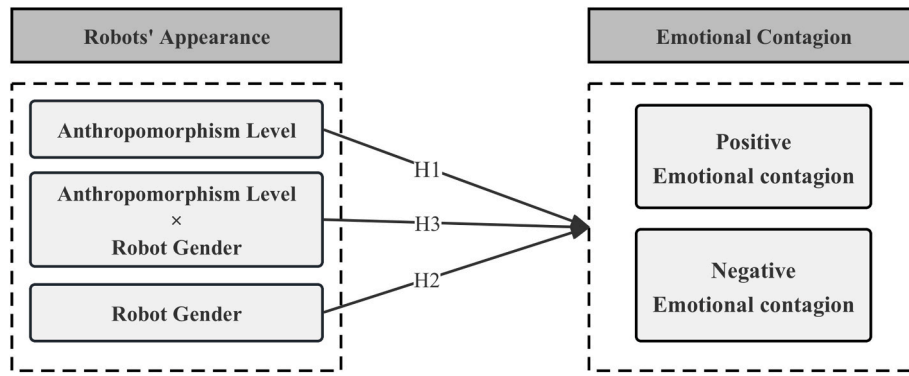


Fig. 1. The proposed theoretical model.

Table 1

Users' visual perception of different anthropomorphic levels of male robots.

	1	2	3
Male Robot Appearance			
Anthropomorphism Score	M = 4.078 SD = 2.168	M = 7.031 SD = 1.776	M = 9.609 SD = 0.908
Visual Gender Perception	75.78 %	89.23 %	100.00 %

Table 2

Users' visual perception of different anthropomorphic levels of female robots.

	1	2	3
Female Robot Appearance			
Anthropomorphism Score	M = 4.078 SD = 2.962	M = 7.328 SD = 2.065	M = 9.453 SD = 0.601
Visual Gender Perception	85.25 %	90.15 %	100.00 %

in emotional contagion is revealed. This literature review emphasizes robots' anthropomorphism and gender as two appearance features that may influence emotional contagion in HRI. Fig. 1 illustrates the theoretical model proposed in this study.

3. Experimental design and methods

3.1. Experiment procedure and participants

This experiment is dedicated to investigating the effects of distinct social robot anthropomorphism and gender on diverse emotional contagion scenarios. The study necessitates selecting three distinct anthropomorphism levels (low, medium, high) for social robots, each encompassing both male and female genders, resulting in six varied social robots with distinctive appearances. Employing a between-groups design for varying robot appearances, the study adopts a within-groups design for varied robot-context interactions.

Each group of participants, after completing an initial emotional questionnaire, engages in viewing positive/negative emotion videos (with randomized sequencing and participants restricted from skipping, pausing, or replaying). Participants subsequently respond to an attention-check question and proceed to evaluate the robot's emotional expression and their emotional involvement. Individuals who have participated in one experimental group are precluded from engaging in other groups.

A total of 187 voluntary participants took part in this study via the Credamo platform, with 186 individuals completing the survey. This

study approved by the ethical committee, strictly adhered to ethical guidelines, ensuring participant autonomy, control, and informed consent. Participants had full authority over their participation, including data access, the right to modify or withdraw consent without negative consequences, and were fully informed about the study's aims, procedures, risks, and benefits. We ensured their anonymity, data confidentiality, voluntary participation with withdrawal options, and took measures to reduce psychological distress, including providing post-experiment support and debriefing. The survey consisted of 52 items. Among these, 6 participants provided erroneous or outlier responses to the emotion contagion videos, and were therefore excluded. The final effective participant count amounted to 180, yielding a response rate of 96.26 %. Participant ages ranged from 20 to 54 years ($M = 31.19$, $SD = 5.21$), and on average, each participant invested 7.58 min ($SD = 2.65$) in completing the survey. Among the participants, 176 reported prior interactions with social robots. All participants exhibited normal vision and hearing, were native Mandarin speakers, possessed sound mental health, and had no prior involvement in similar studies. Importantly, none of the participants reported feelings of strangeness, mystery, or fear during the experiment, thus indicating an absence of the uncanny valley effect. The post-experiment questionnaire affirmed complete participant comprehension of the simulated scenarios and verified that any emotional fluctuations were attributed to the simulated interactions with social robots.



Fig. 2. Opening scene of HRI video (e.g., moderately anthropomorphized male robot).

3.2. The manipulation of social robot appearance

For assessing the degree of anthropomorphism, we adopted the approach by Phillips et al., which gauges robot anthropomorphism across 19 dimensions, encompassing human-like attributes (e.g., glasses, skin, hair) as well as non-human traits (e.g., wheels, tracks) (Phillips, Xuan, Ullman, & Malle, 2018). A cohort of 60 participants in this study evaluated the robots using a 10-point scale, spanning from “not human-like” to “completely human-like,” adhering to predefined evaluation criteria. Notably, the low anthropomorphism robot selected was DD02 by Beiens ($M = 4.078$, $SD = 1.595$), the medium anthropomorphism robot was ROBOT K30 by SMART ROBOT ($M = 7.180$, $SD =$

1.389), and for high anthropomorphism, live actors were employed ($M = 9.531$, $SD = 0.869$) to avert the uncanny valley effect.

To imbue social robots with gender attributes while maintaining consistent anthropomorphism levels, we introduced gender-specific traits. These visual gender cues acted as supplementary indicators to complement gender explanations provided through video subtitles and robot-associated voice cues. As detailed in Tables 1 and 2, preliminary findings underscore that the perception accuracy of visual cues surpassed 75 % for all robots.

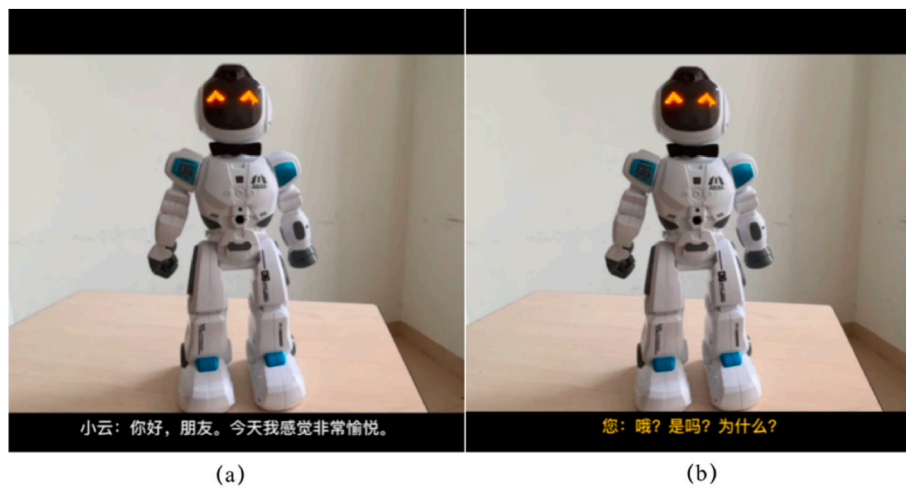
3.3. Emotional contagion scenario design

In this experiment, we designed emotion contagion dialogue videos of two different emotions for HRI scenarios. In the scenario of positive emotion, the robot showed positive emotions due to receiving affirmation for its abilities. In the scenario of negative emotion, the robot expressed negative emotions because of poor task performance. Participants were first shown a video introducing the experimental context, robot identity, and gender through static images, text, and voiceover (as in Fig. 2). Male and female social robots were named “Xiao Yun.” The video began with a subtitle and voiceover stating, “The upcoming video records your interaction with the male/female social robot Xiao Yun.” The video showed interactions between the robot and humans under positive or negative emotions, with the order randomized. Participant speech was in yellow subtitles, and robot speech was in white (as in Fig. 3). The first-person perspective video aided participants’ role engagement in HRI.

Considering the experimenters’ language environment, emotional contagion video scripts were adapted from the Chinese EmoBank (Lee, Li, & Yu, 2022) and modified for the HRI context. Robots of the same gender shared voices to prevent voice cues influencing gender perception and emotional contagion.

3.4. Measure of emotional contagion

In previous research, researchers have often employed self-report measures to assess the emotional contagion effect. For instance, in (Matsui & Yamada, 2016), participants were asked to self-report their feelings of happiness before and after human-robot interaction. In (Xu, et al., 2015), participants completed Self-Assessment Manikin (SAM) affect self-reports and post-experiment questionnaires. In (Stock, 2016),



(a) Video Frame Example of Robot “Xiao Yun” Speaking
(b) Video Frame Example of User Speaking

Fig. 3. Content example of HRI video (e.g., moderately anthropomorphized male robot).

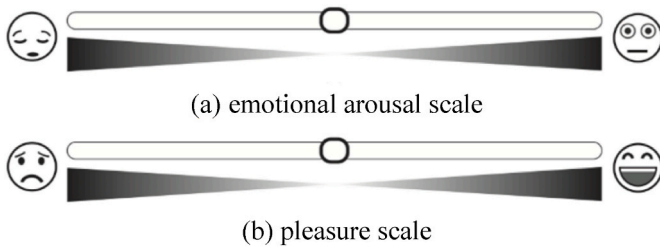


Fig. 4. Emotional slider scale developed by Betella and Verschure (2016)

participants' self-reported measures of pleasant affect were compared with parallel measures taken in the latter part of the experiment.

Similarly, in this study, we utilize the "pre-and-post emotional delta" to represent the emotional contagion effect. This measurement approach has been chosen because it effectively captures changes in individuals' emotional states following exposure to emotional stimuli. Building upon the work of Betella and Verschure (2016) (see Fig. 4), this study retained the measurement scale for pleasure (Scale b), to align with the experimental context and enhance the readability of the questionnaire.

4. Experimental results

4.1. Validity analysis

The Pearson correlation between participants' perception of the robot's emotions and their emotional changes using the correlation

Table 3
Effectiveness testing results for emotional contagion.

Robot Anthropomorphism Level	Robot Gender	Pearson Correlation Coefficient	P
Low	Male	0.778	<0.001
	Female	0.735	<0.001
Middle	Male	0.712	<0.001
	Female	0.756	<0.001
High	Male	0.731	<0.001
	Female	0.764	<0.001

Table 4
Descriptive statistics results.

Experiment Conditions	Robot Anthropomorphism Level	Robot Gender	Mean	Standard Deviation	N
Positive Emotion	Low	Male	1.77	2.609	30
		Female	1.14	2.15	30
	Middle	Male	1.7	2.351	30
		Female	0.96	1.732	30
	High	Male	0.9	1.291	30
		Female	1.5	1.852	30
Negative Emotion	Low	Male	4.33	2.171	30
		Female	3.17	2.253	30
	Middle	Male	3.67	2.708	30
		Female	3.46	2.236	30
	High	Male	3.07	2.477	30
		Female	3.67	2.758	30

Table 5
Analysis of variance results for the effects of social robots' anthropomorphism and gender on emotional contagion in HRI.

Independent Variable	Dependent Variable	Mean Square	F	p
Robot Anthropomorphism	Positive Emotional Contagion	5.881	1.211	0.308
	Negative Emotional Contagion	0.216	0.035	0.998
Robot Gender	Positive Emotional Contagion	23.719	4.885	0.028**
	Negative Emotional Contagion	0.094	0.015	0.902
Interaction Term	Positive Emotional Contagion	14.924	3.074	0.049**
	Negative Emotional Contagion	1.809	0.292	0.747

coefficient. Table 3 demonstrates a significant correlation between participants' perception of the robot's emotions and their emotional changes, indicating the presence of emotional contagion.

4.2. Testing the effects of robot appearance on emotional contagion

We conducted descriptive statistics and multifactor analysis of variance (ANOVA) to test the hypotheses derived from the literature. The relevant data passed the homogeneity of variance test (positive emotional changes: $p = 0.09 > 0.05$, negative emotional changes: $p = 0.384 > 0.05$). Table 4 presents the descriptive statistics of the data.

Table 5 displays the ANOVA results of the effects of anthropomorphism level and gender of social robots on emotional contagion during HRI, indicating whether hypotheses are supported.

Under the experiment condition of positive emotion, ANOVA results show that social robots' anthropomorphism has no significant effect on emotional contagion ($F = 1.211$, $p = 0.308 > 0.05$), social robots' gender has a significant impact on emotional contagion ($F = 4.885$, $p = 0.028 < 0.05$), the interaction effect of social robots' anthropomorphism and gender also has a significant impact on the contagion of positive emotion ($F = 3.074$, $p = 0.049 < 0.05$), thus supporting H2a and H3a, and not supporting H1a.

For negative emotions, neither social robots' anthropomorphism nor gender has a significant impact on emotional contagion, and still, there is no significant interaction effect on emotional contagion, thus not supporting H1b, H2b, and H3b.

Fig. 5 describes the impact of social robots' gender on the contagion of positive emotion. It can be observed that users exhibit stronger changes of positive emotion when interacting with male compared to female social robots. In other words, male social robots exert a stronger contagion of positive emotion on users in HRI compared to female robots.

Based on the significant interaction effects observed between social robot anthropomorphism and gender on emotional contagion, we further conducted simple effects tests. The results showed that in the case of a high anthropomorphism level of the social robot, the social robot's gender has a significant impact on the contagion of positive emotion ($p = 0.04 < 0.05$). This finding provides further support for hypothesis H3.

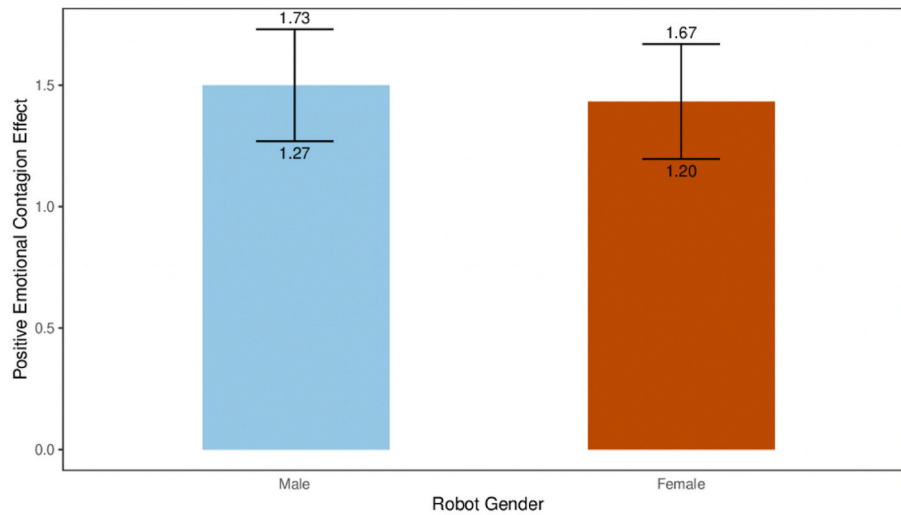


Fig. 5. Effects of robots' gender on the contagion of positive emotion.

Table 6

Results of simple effects analysis.

Robot Anthropomorphism Level	Robot Gender	SD	SE	t	p
Low	Male-	0.63	0.534	1.18	0.24
	Female				
Middle	Male-	0.74	0.539	1.37	0.17
	Female				
High	Male-	-4.1	2.058	-1.99	0.04**
	Female				

The simple effect analysis (see Table 6) means that, under the high anthropomorphism level of social robots, users experienced greater changes of positive emotion when interacting with female social robots compared to male social robots. In other words, when social robots are at a higher anthropomorphism level, female social robots have a stronger contagion of positive emotion on users compared to male social robots. Fig. 6 further describes the effects of this interaction.

5. Conclusion and discussion

5.1. Conclusions

This study investigates the impact of social robots' appearance on emotional contagion in HRI, specifically examining the influence of their anthropomorphism and gender, as well as their interacting effects. The experimental findings revealed that social robots' gender significantly affects the contagion of positive emotions in HRI, but not negative emotions. In contrast, social robots' anthropomorphism does not significantly influence the contagion of either positive or negative emotions. Additionally, the combined effects of social robots' anthropomorphism and gender significantly impact the contagion of positive emotion. Fig. 7 provides an overview of the hypotheses in the conceptual model and the corresponding test results.

5.2. Theoretical contributions

This study extends the knowledge of emotional contagion in the field of human-robot interaction. First, this study investigates the effects of social robots' appearances on the contagion of positive and negative emotions. Although previous studies have investigated emotional

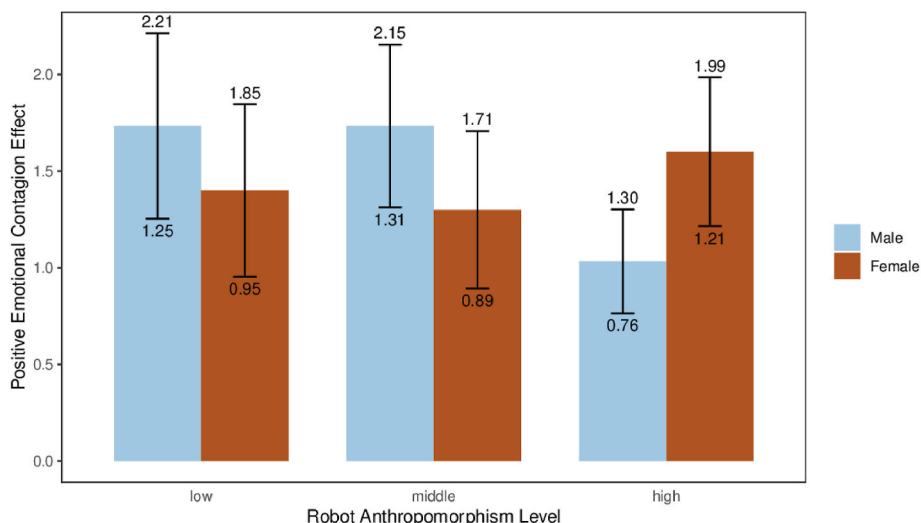


Fig. 6. The interacting effect of Social Robots' Anthropomorphism and Gender on the Contagion of Positive Emotion.

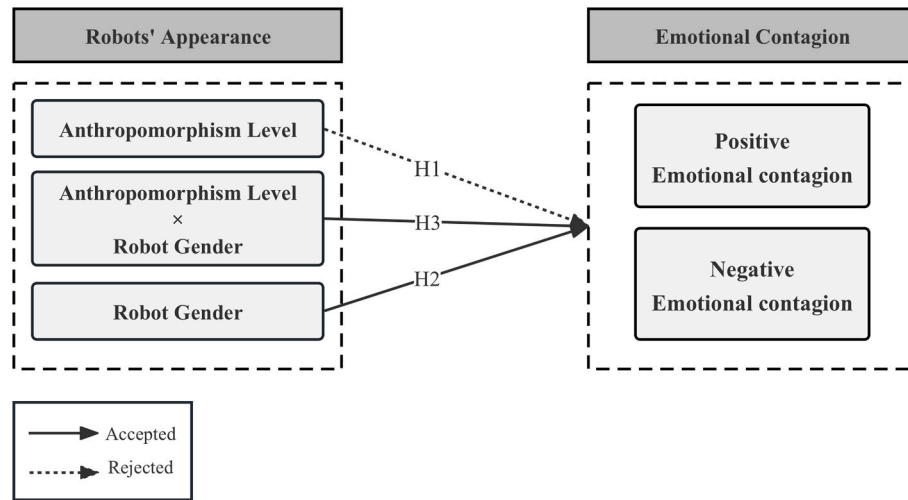


Fig. 7. Conceptual model of the validated impact of social robot anthropomorphism level and gender on emotional contagion in HRI.

contagion during HRI (Casso, et al., 2022; Chuah & Yu, 2021; Matsui & Yamada, 2016, 2019; Otterbring & Lu, 2018; Stock, 2016; Xu et al., 2015), rare studies focus on the effects of robots' appearance or take both positive and negative emotions as the consequences. This study provides a new perspective on the factors influencing on social robots' emotional contagion.

Secondly, this study's findings about the effect of robots' gender on emotional contagion contribute to the existing literature in the field. Previous studies have reported differences in the impact of humans' different gender faces on human emotional responses, where pictures of happy females can evoke more happiness, and pictures of sad females can evoke more sadness (Wild, Erb, & Bartels, 2001). However, this study investigated the effect of robots' gender on emotional contagion, indicating that social robots' gender plays a role in the contagion of positive emotion during human-robot interaction. Besides, the significant effects of robot gender on emotional contagion highlight the relevance of gender stereotypes in the process of emotional contagion during human-robot interaction, which may contribute empirical insights to the field of gender studies in robotics.

Third, this study is the first to consider the effects of robots' anthropomorphism and gender simultaneously on emotional contagion during HRI. In previous studies (Matsui & Yamada, 2019; Volante et al., 2016), researchers have discussed the effects of virtual human images with different levels of anthropomorphism on emotional contagion. The experimental results showed that highly anthropomorphic virtual human images exert stronger emotional contagion on humans. However, the findings of this study show that robots' emotional contagion of positive or negative emotions under robot images of different genders and different levels of anthropomorphism may not be simply linear. The effect of social robots' gender on the contagion of positive emotion is amplified by the level of robots' anthropomorphism. Thus, this study contributes to the literature by investigating more factors simultaneously rather than on a single factor.

Future direction: These different conclusions from previous studies found in our experiments suggest that the factors influencing robots' emotional contagion in human-computer interaction still need to be reassessed such as cultural differences, experimental contexts, and image selection.

5.3. Practical implications

The practical implications of this research extend to both the design and utilization of social robots, offering insights in two key areas: the design of social robots and their selection based on the requirements of emotional contagion.

First, stakeholders involved in selecting emotional companion social robots should consider the gender representation of the robots. This study reveals a significant difference in the contagion of positive emotions between male and female robots in human-robot interaction and no significant difference in negative emotions. The strategic approach of selecting robot appearance aligns with the evident effect of robot gender on emotional engagement. Specifically, the experimental findings of this study indicate that, in the context of positive emotions, highly anthropomorphic male robots exhibit a stronger emotional contagion towards users than female robots in human-robot interaction. Conversely, for low anthropomorphic robots, this effect is reversed. Thus, when stakeholders need to transmit positive emotions through social robots, it might be a wiser decision to select social robots of different gender appearances based on the level of anthropomorphism of the robot.

Second, comprehending the interplay of social robot anthropomorphism and gender in emotional contagion provides a valuable resource for refining robot design. Researchers conducting experiments centered on emotional contagion within human-robot interaction can strategically select robot designs that align with the emotional categories relevant to their investigations. Simultaneously, these findings offer insights for researchers in interpreting their experiments. This study shows that the effects of social robots' anthropomorphism on emotional contagion are influenced by robots' gender, and the effects of social robots' gender on emotional contagion are influenced by robots' anthropomorphism. This knowledge enables the optimization of robots to enhance their efficacy in emotional communication and companionship, thereby addressing users' specific needs, particularly in sectors such as health-care, education, and customized robotic companion services. In the context of robot selection, manufacturers and businesses have opportunities to harness specific robots tailored to the desired emotional outcomes. The findings of this study provide business implications on how to select and deploy robots that align with the intended emotional atmosphere or context.

5.4. Limitations and future research avenues

This study has certain limitations. One limitation pertains to the use of video-based conversation scenarios, which may be perceived as less realistic and consequently may restrict the generalizability of participants' responses and the richness of their emotional expressions. Conducting field studies involving social robots with varying genders and levels of anthropomorphism in real-life settings could be considered a promising alternative approach to assess users' emotional contagion effects more authentically.

The second limitation is that the experimental design of this study

did not consider the Uncanny Valley effect, although the three selected humanoid forms used in this study only provide minimal interaction and do not reach the level necessary to induce the Uncanny Valley effect (Kim, Schmitt, & Thalmann, 2019). The Uncanny Valley effect, which describes a drop in human acceptance and an increase in discomfort as robots become too human-like, was not considered in our study's design (Mori, MacDorman, & Kageki, 2012). It is important to note that the Uncanny Valley effect occurs in environments that involve a high level of anthropomorphism and various sensory stimuli, including auditory, visual, and tactile cues (T. Nomura & Kanda, 2016). Factors like low-level deviations in humanoid robots and facial arrangement violations contribute to this effect (MacDorman, Green, Ho, & Koch, 2009; Mori et al., 2012; Seyama & Nagayama, 2007). Mori et al. (2012) suggested the effect is more pronounced for dynamic stimuli. The anthropomorphic design of social robots can challenge accurate interaction expectations in HRI (Dereshev, Kirk, Matsumura, & Maeda, 2019). Users may develop incorrect expectations of the robot's cognition and social capabilities during prolonged interactions (Dautenhahn, 2004). Thus, in future experiments, the design should take into consideration the Uncanny Valley effect.

Another limitation of this study is its omission of an in-depth analysis of the long-term exposure and impact of emotional contagion in human-robot interaction. Notably, sustained interactions with social robots could potentially alter a user's level of emotional investment, trust, and dependency on these robots (Van Maris, et al., 2020). Future research should thus explore the potential long-term effects of such interactions on user emotions and behavior.

Additionally, our study's focus on the role of robot appearance in emotional contagion brings to the forefront the critical need for addressing user autonomy and control in human-robot interactions. Integrating considerations of user autonomy into the design principles of social robots is essential. This integration not only enriches the user experience but also confronts ethical challenges, such as the potential for manipulation and the imperative to respect user autonomy (Maroto-Gómez, et al., 2023; Pollmann, Loh, Fronemann, & Ziegler, 2023). Future work should include the development and application of methodologies in experimental settings that evaluate and give precedence to user autonomy and control during interactions with social robots.

Furthermore, in order to gain a comprehensive understanding of emotional contagion in human-robot interaction, future investigations could explore additional user-level factors, such as personality traits, attitudes towards robots, and prior experiences with social robots, to provide more nuanced insights into the complex dynamics of emotional contagion.

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CRediT authorship contribution statement

Wenjing Yang: Conceptualization, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **Yunhui Xie:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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